Hidden Distributional Sources of Group Fairness in Collaborative Filtering

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Abstract

Fairness evaluations of recommender systems commonly attribute differences in error between user groups to group representation, sensitive attributes, or stereotypical preferences. We argue that this view is incomplete for collaborative filtering: the distribution of observed ratings also determines what a model learns and which errors are visible at evaluation time. We introduce an experimental framework for isolating hidden distributional factors, like user activity, rating eccentricity and other psychosociological traits, and to discompose their effects on perceived group unfairness, both due to inductive model biases or disparate distributions. Across our experiments, measured group unfairness can be decomposed and understood by employing our proposed method, helping explain how and why collaborative filtering models favor one group over another. These results challenge explanations of recommender group unfairness and motivate distribution-aware fairness analyses like our proposal.

1 Introduction

Recommender systems learn from interactions that are neither uniformly distributed nor independent of the users and items producing them. A user with many observed interactions supplies a different training signal from a user with only a few. Users and items also differ in how typical their observed interactions are. These properties are usually treated as data diagnostics, yet they can become ingredients of model disparity: they influence the representations learned by a collaborative model and determine which examples dominate an error metric.

This issue is particularly important for group fairness. Existing analyses often organize fairness around protected-group representation, popularity, or the possibility that a model reproduces social stereotypes [6, 7, 4, 13]. Those mechanisms matter, but a recommender can show a group error gap even when the sensitive attribute is not used by the model and even when group membership is assigned from artifical groups not representing real demographic factors.

The goal of this work is to provide a controlled way to characterize distributional sources of group unfairness in Recommender systems. We consider not only known aspects of dyadic data like group representation and psycho-sociological criteria, but also hidden factors, namely user activity and rating eccentricity, which respectively describe how much concentrated the interaction population is, and how far an observed rating departs from local user-item behavior. Eccentricity, in particular, is motivated by a phenomenon recently described by Paz-Ruza et al. [14], who show that non-uniform observed-value distributions can make collaborative filtering models unassumingly favour "average" cases and perform disproportionately poorly on unexpected or "eccentric" interactions.

Our experimental question is not knowing what groups are more or less advantaged, but whether we can precisely attribute the perceived group unfairness of collaborative filtering models to a combination of known and hidden distributional factors disregarded in existing literature and practice. We make four main contributions, summarized on Figure **??**:

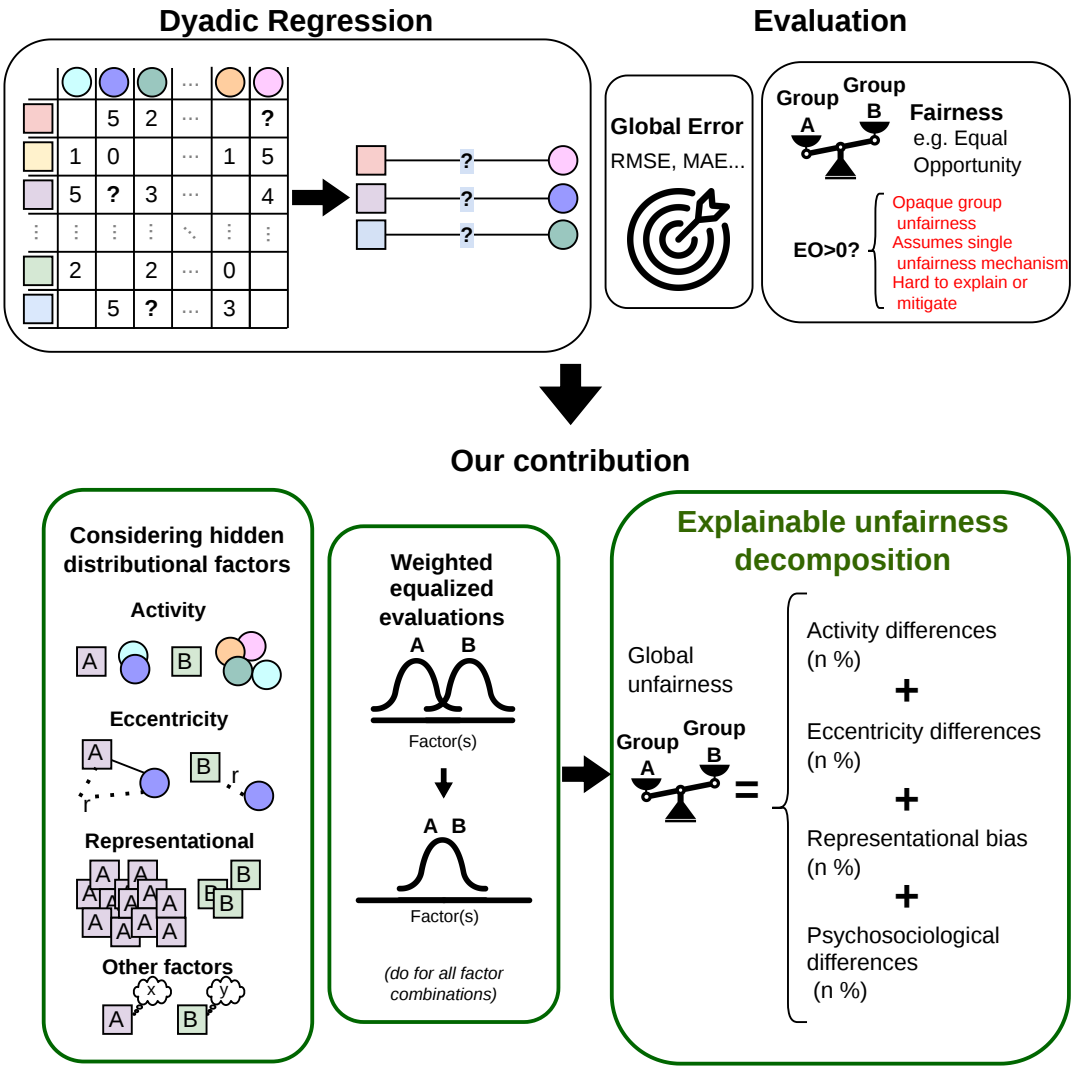


Figure 1: Summary and rationale of this work's main contributions.

1. We formulate activity and eccentricity as additional hidden distributional bias factors that can be analyzed independently of the demographic meaning of a group.

2. We define a reproducible procedure for additive decomposition of the observed group unfairness of models into bias arising from eccentricity, activity, representational and other psychosociological factors that may be disparate between groups.

3. We experiment in real-world collaborative filtering datasets considering natural, demographic group constructions as well as controlled, induced splits. Our results demonstrate the effectiveness of our method to decompose group unfairness sources.

4. We prove the influence of activity and eccentricity disparities among groups in recommender fairness, challenging previous assumptions that only consider representational or psychosociological biases.

2 Background and Related Work

This section introduces the fairness perspective, collaborative-filtering notation, and prior work that motivates our proposal for a decompositional algorithm of group unfairness treating activity and eccentricity as possible distributional factors.

2.1 Collaborative filtering task and notation

Collaborative filtering predicts a user's preference for an item from a sparse collection of observed user-item interactions. Let $\mathcal{U}$ and $\mathcal{I}$ denote users and items, and let $R \subseteq \mathcal{U} \times \mathcal{I} \times \mathbb{R}$ be the observed interaction set. An element $(u, i, r_{ui}) \in R$ records that user u gave item i the rating r_{ui}. The user and item neighborhoods are $R_u = \{(i, r_{ui}) : (u, i, r_{ui}) \in R\}$ and $R_i = \{(u, r_{ui}) : (u, i, r_{ui}) \in R\}$. Their empirical means are

$$\bar{r}_u = \frac{1}{|R_u|} \sum_{(i, r_{ui}) \in R_u} r_{ui}, \qquad \bar{r}_i = \frac{1}{|R_i|} \sum_{(u, r_{ui}) \in R_i} r_{ui}. \tag{1}$$

Models operating in collaborative filtering tasks are of the form $r_{ui} = \phi(u, i, \cdots)$. Because the model ϕ learns from R, the composition of the observed interactions affects both the evidence available for each representation and the errors measured on later interactions.

2.2 Fairness in recommender systems

Fairness is not a single property in recommendation. It can concern users, items, providers, exposure, utility, or the relation between predictions and outcomes. Multisided formulations emphasize that different stakeholders can have incompatible objectives [4], while ranking work studies exposure and attention rather than only prediction error [15, 3]. Surveys have consequently distinguished individual, group, consumer, provider, and multistakeholder notions of fairness [7, 13]; in this work, we work on the popular subproblem of group fairness in direct ranking prediction tasks.

Representation is often considered a central source of algorithmic disparity in Machine Learning. In Recommender systems, minority groups may have fewer observations, less coverage, or systematically different interaction patterns. In movie recommender systems, Gonzalez et al. [8] analyze bias and unfairness in collaborative filtering and attributed representational bias to be the culprit of observed disparities.

These accounts, however they do not imply that representation is the only mechanism. In matrix factorization, observation counts affect the amount of evidence available for user and item embeddings. The values of those observations also affect the loss landscape, both in optimization and in evaluation. A group can therefore be disadvantaged by the kinds of ratings it contributes, not only by the number of users or ratings it contributes or a psychosociological difference in their behaviour.

Further than evaluation, interventions to mitigate group unfairness include a myriad of pre-processing, in-processing, and post-processing approaches, as well as constraints on ranking exposure or group representation [12, 9, 5]. These lines of work establish important group-level and stakeholder-level and ethical positions to systematically reduce the overall group unfairness of collaborative filtering models, but they do not by themselves explain what is the actual source of such biases, and do not investigate whether the internal distribution of collaborative filtering evidence creates unequal prediction errors.

2.3 Activity, popularity, and eccentricity

Here, we describe different distributional factors that may affect group fairness in collaborative filtering:

- User activity and item popularity are the observation counts

$$a_u = |R_u|, \qquad p_i = |R_i|. \tag{2}$$

  Activity measures how much evidence is available for a user representation; popularity measures how frequently an item appears in the observed interaction graph. Item popularity bias has been extensively studied in learning-to-rank case, where items with large p_i in

binary interaction graphs get disproportionately recommended at the expense of long-tail items [11, 1, 2]. In rating prediction tasks, item popularity is part of the background mechanism that can shape learned item representations, not affecting directly user group unfairness, while disparate user activity distributions may affect the quality of learning of the preferences for users of different groups, regardless of their overall group representation.

- For each observed interaction, define its local eccentricity by

$$\epsilon_{ui} = \left| r_{ui} - \frac{\bar{r}_u + \bar{r}_i}{2} \right|. \tag{3}$$

The corresponding user-level eccentricity is the mean over that user's observed interactions,

$$e_u = \frac{1}{|R_u|} \sum_{(i, r_{ui}) \in R_u} \epsilon_{ui}. \tag{4}$$

Thus, ϵ_{ui} describes how atypical an individual observation is relative to the expected for the dyad.

Paz-Ruza et al. [14] formalize eccentricity bias in dyadic regression and introduce a metric to characterize overall model bias due to disproportionately punishing eccentric cases. Their analysis demonstrates why global RMSE and MAE can hide structured errors, but do not translate it to individual or group fairness. In our work, eccentricity is not itself a group definition; it is a candidate hidden factor that may help explain why two groups receive different prediction errors.

3 Proposal

This section defines the proposed diagnostic pipeline, consisting of evaluation- and training-time interventions on the measurement of group fairness biases in collaborative filtering. In particular, we propose to isolate and decompose how different individual distributional factors of the training and test data (e.g. user activity, rating eccentricity, group representation and other psychosociological factors) affect intrinsic model biases and group fairness metrics in the final evaluation.

3.1 Group constructions

Group membership is represented by a user-level label $g_u \in \{A, B\}$, so every interaction by a user inherits that user's group. The proposal is agnostic to how the two labels are obtained: they may denote observed groups, controlled induced populations, or another user-level partition. The precise group constructions and their parameters are implementation choices reported in Section 4.

For a feature x and population parameters (μ_A, σ_A) and (μ_B, σ_B), the induced group probability is

$$P(A \mid x) = \frac{\mathcal{N}(x; \mu_A, \sigma_A)}{\mathcal{N}(x; \mu_A, \sigma_A) + \mathcal{N}(x; \mu_B, \sigma_B)}. \tag{5}$$

The label is sampled once per user. This construction deliberately separates a distributional factor from a real-world protected attribute; the same formal definition also covers any other user-level group assignment.

3.2 Test-side weighting and EO decomposition

The test-side procedure attributes the model's overall observed group error difference to hidden distributional factors. Holding the predictor and its training data fixed, it constructs counterfactual evaluation distributions in which activity and eccentricity are successively aligned

between groups. The quantities needed for its decomposition are obtained through the weighted evaluations performed on four different conditions:

- $\mathcal{C}_0$: no weighting;
- $\mathcal{C}_a$: activity weighting
- $\mathcal{C}_e$: eccentricity weighting
- $\mathcal{C}_{ae}$: joint activity–eccentricity weighting

Algorithm 1 produces the evaluation quantities used for attribution in six steps:

1. It trains the checkpoint and restricts evaluation to test ratings whose users and items are represented in training (lines 1–2).
2. It computes the training-set reference means and the pooled activity and eccentricity bins (lines 3–8).
3. It selects one of the four named conditions and, when required, assigns each eligible rating to its activity, eccentricity, or joint stratum.
4. For each shared stratum, it counts the two groups, converts the counts to group proportions, and constructs the normalized common target (lines 23–24).
5. It assigns importance weights, giving no weight to group-exclusive strata, and evaluates the unchanged checkpoint with those weights (lines 26–29).
6. It records the condition-specific metrics for the EO decomposition (line 32).

Algorithm 1 Test-set weighting and fixed-checkpoint evaluation

Require: Training set T, test set V; quantile counts Q_a, Q_e
Ensure: Predictions and metrics for conditions $\mathcal{C} = \{\mathcal{C}_0, \mathcal{C}_a, \mathcal{C}_e, \mathcal{C}_{ae}\}$

Train one model M on T and save its checkpoint
$E \leftarrow \{v \in V : u(v) \in U(T) \ \wedge \ i(v) \in I(T)\}$
Compute $\bar{r}_u$ and $\bar{r}_i$ using T only
for all $v = (u, i, r, g) \in E$ **do**
    $a(v) \leftarrow |\{r' \in T : u(r') = u\}|$
    $e(v) \leftarrow |r - (\bar{r}_u + \bar{r}_i)/2|$
end for
Compute pooled quantile bins $b_a(v)$ and $b_e(v)$ from E
for all $c \in \mathcal{C}$ **do**
    if $c = \mathcal{C}_0$ **then**
        Set $w(v) \leftarrow 1$ for all $v \in E$
    else
        if $c = \mathcal{C}_a$ **then**
            $s_c(v) \leftarrow b_a(v)$
        else if $c = \mathcal{C}_e$ **then**
            $s_c(v) \leftarrow b_e(v)$
        else
            $s_c(v) \leftarrow (b_a(v), b_e(v))$
        end if
        for all s containing ratings from both groups **do**
            $n_{s,g} \leftarrow |\{v \in E : s_c(v) = s, \ g(v) = g\}|$, $g \in \{A, B\}$
        end for
        $p_{s,g} \leftarrow n_{s,g} / \sum_t n_{t,g}$
        $q_s \leftarrow \min(p_{s,A}, p_{s,B})$ and $q_s \leftarrow q_s / \sum_t q_t$
        for all $v \in E$ with group g and stratum $s_c(v) = s$ **do**
            $w(v) \leftarrow q_s / p_{s,g}$ if s is shared; otherwise $w(v) \leftarrow 0$
        end for
    end if
    Evaluate the unchanged checkpoint of M on E using w
    Compute weighted group RMSEs and EO_c
end for
Record the metrics for each weighting condition

3.2.1 Additive EO decomposition

Let RMSE_A and RMSE_B be the group-specific root mean squared errors. We use the signed error difference

$$\mathrm{EO} = \mathrm{RMSE}_B - \mathrm{RMSE}_A. \tag{6}$$

Positive EO means that group A has lower prediction error and is therefore favored by the predictor; negative EO favors group B. The unweighted evaluation gives the overall observed gap, denoted by EO_0. The three reweighted evaluations give EO_a, EO_e, and EO_{ae} for activity, eccentricity, and joint activity–eccentricity alignment, respectively.

We use the additive Shapley decomposition

$$\Phi_a = \tfrac{1}{2}[(\mathrm{EO}_0 - \mathrm{EO}_a) + (\mathrm{EO}_e - \mathrm{EO}_{ae})], \tag{7}$$
$$\Phi_e = \tfrac{1}{2}[(\mathrm{EO}_0 - \mathrm{EO}_e) + (\mathrm{EO}_a - \mathrm{EO}_{ae})], \tag{8}$$
$$\Phi_o = \mathrm{EO}_{ae}, \tag{9}$$
$$\mathrm{EO}_0 = \Phi_a + \Phi_e + \Phi_o. \tag{10}$$

Here Φ_a is the activity contribution, Φ_e the eccentricity contribution, and Φ_o the residual attributable to other factors, including representation and psychosociological differences not controlled by these constructions.

3.3 Train-side extension

The train-side procedure attributes changes in the learned model's EO to the composition of its interaction population. It constructs two training conditions from complete users:

- $\mathcal{T}_b$: a fully balanced user-closed training subset;
- $\mathcal{T}_r$: a resized user-closed control with the same rating count as $\mathcal{T}_b$ and no balancing criterion.

Both conditions preserve whole-user interactions. The first changes the composition of the population by reducing stratum-level group imbalance; the second isolates the effect of the resulting training-set size.

Algorithm 2 constructs $\mathcal{T}_b$:

1. It computes user features, group labels, and joint activity– eccentricity strata (lines 1–2).
2. It repeatedly identifies the most imbalanced shared stratum and removes one complete user from its overrepresented group (lines 4–11).
3. It returns the user-closed balanced subset and its rating count (lines 15–16).

Algorithm 2 Construction of the fully balanced training subset

Require: Ratings R; quantile counts Q_a, Q_e; tolerance τ
Ensure: Training condition $\mathcal{T}_b$ and reference size N_{ref}
1: Compute (a_u, e_u, g_u) for every $u \in \mathcal{U}$
2: Compute $b_a(u)$ and $b_e(u)$; set $s(u) \leftarrow (b_a(u), b_e(u))$
3: $\mathcal{U}' \leftarrow \mathcal{U}$ and $\rho^\star \leftarrow 1$
4: Compute $n_{s,g}(\mathcal{U}')$ for all strata and groups
5: $\rho_s(\mathcal{U}') \leftarrow n_{s,A}(\mathcal{U}')/n_{s,B}(\mathcal{U}')$ for shared strata
6: $d_s(\mathcal{U}') \leftarrow \max\{\rho_s(\mathcal{U}')/\rho^\star, \rho^\star/\rho_s(\mathcal{U}')\}$
7: **while** $\max_{s\in\mathcal{S}_{\text{shared}}(\mathcal{U}')} d_s(\mathcal{U}') > \tau$ **do**
8: $\quad s^\star \leftarrow \arg\max_{s\in\mathcal{S}_{\text{shared}}(\mathcal{U}')} d_s(\mathcal{U}')$
9: $\quad g^\star \leftarrow \arg\max_{g\in\{A,B\}} n_{s^\star,g}(\mathcal{U}')$
10: $\quad C \leftarrow \{u \in \mathcal{U}' : s(u) = s^\star,\ g_u = g^\star\}$
11: $\quad$ Randomly permute C and select its first user $u^\star$
12: $\quad \mathcal{U}' \leftarrow \mathcal{U}' \setminus \{u^\star\}$
13: $\quad$ Update $n_{s,g}(\mathcal{U}')$, $\rho_s(\mathcal{U}')$, and $d_s(\mathcal{U}')$
14: **end while**
15: $R_{\text{bal}} \leftarrow R(\mathcal{U}')$
16: $N_{\text{ref}} \leftarrow |R_{\text{bal}}|$
17: $\mathcal{T}_b \leftarrow R_{\text{bal}}$
18: **return** $(\mathcal{T}_b, N_{\text{ref}})$

Algorithm 3 constructs $\mathcal{T}_r$ by matching the rating count returned for $\mathcal{T}_b$:

1. It computes the number of ratings to remove and the reachable sums of complete-user activity counts (lines 1–5).
2. It verifies that the target count is achievable by whole-user removal and reconstructs one valid removal set (lines 8–13).
3. It returns the user-closed resized control $\mathcal{T}_r$ (line 16).

Algorithm 3 Exact whole-user resizing without balancing

Require: Ratings R; reference size N_{ref}
Ensure: Training condition $\mathcal{T}_r$
Compute a_u for every $u \in \mathcal{U}$ and set $K \leftarrow |R| - N_{\text{ref}}$
Randomly permute users to obtain $(u_1, \ldots, u_{|\mathcal{U}|})$
Initialize reachable sums $S_0 \leftarrow \{0\}$
for $j = 1$ to $|\mathcal{U}|$ **do**
    $S_j \leftarrow S_{j-1} \cup \{x + a_{u_j} : x \in S_{j-1},\ x + a_{u_j} \leq K\}$
end for
if $K \notin S_{|\mathcal{U}|}$ **then**
    raise "No exact complete-user resizing exists"
end if
$q \leftarrow K$ and $\mathcal{V} \leftarrow \varnothing$
for $j = |\mathcal{U}|$ downto 1 **do**
    if $q \notin S_{j-1}$ **then**
        $\mathcal{V} \leftarrow \mathcal{V} \cup \{u_j\}$ and $q \leftarrow q - a_{u_j}$
    end if
end for
return $R(\mathcal{U} \setminus \mathcal{V})$ as $\mathcal{T}_r$

4 Experimental Configuration

Data and group constructions All experiments use the MovieLens 1M dataset [10], with 1,000,209 ratings from 6,040 users on 3,706 items. The observed interaction set R introduced in Section 2 contains user–item ratings. The chosen group constructions are:

- **Gender:** group A contains male users and group B female users.
- **Age groups:** group A contains users older than 45 and group B the remaining users.
- **Induced activity:** equal-prior Normal populations with means 300 and 400 ratings and standard deviation 200.
- **Induced eccentricity:** equal-prior Normal populations with means 0.65 and 0.85 and standard deviation 0.15.

For both induced constructions, membership is sampled once per user from the group assignment probability and inherited by all of that user's interactions. Group labels define evaluation populations only; they are not input features of the predictor.

Models and baselines The primary predictor is a matrix-factorization dot-product model with embedding dimension 256 and a maximum of 50 training epochs. Early stopping with 5-epoch patience selects the checkpoint used for evaluation. As baselines, we consider a uniform random predictor and the mean of the user's and item's training ratings.

4.1 Experimental protocol and implementation

The evaluation protocol uses shuffled ten-fold cross-validation, with random seed 727 plus the fold index. The train- and test-side interventions employ 10 and 50 quantile bins respectively to represent hidden distributional factors (activity and/or eccentricity). The fully balanced training condition uses a tolerance of $\tau = 1.05$ for the maximum allowed stratum-level imbalance. The resized control is constructed to match the rating count of the fully balanced condition. The code, experiment scripts, generated CSV files, and plotting resources are available at https://github.com/Kominaru/fair-recommenders-evaluation.

5 Results

This section reports the empirical attribution of group error differences and the complementary effect of changing the training population. The test-side analysis attributes the overall EO through weighted evaluation distributions; the train-side analysis shows how population composition affects the learned model.

5.1 Test-side EO attribution

The proposed pipeline evaluates the unweighted reference distribution and three weighted evaluation distributions. The resulting quantities decompose the observed signed EO. Table 1 reports the fold-averaged EO and its activity, eccentricity, and other contributions for matrix factorization and the two baselines. The contributions reconstruct the global EO to numerical precision.

Table 1: Signed EO decomposition across ten folds. Positive EO favors group A. Values are means ± standard deviations.

Construction	Predictor	Global EO	Decomposition of global EO		
			Activity	Eccentricity	Other
Gender	MF	0.045 ± 0.005	0.005 ± 0.001	0.015 ± 0.004	0.026 ± 0.002
	random uniform	0.007 ± 0.007	0.007 ± 0.002	0.008 ± 0.004	-0.008 ± 0.007
	user item average	0.021 ± 0.006	0.001 ± 0.001	0.018 ± 0.006	0.002 ± 0.001
Age	MF	0.018 ± 0.008	-0.009 ± 0.002	0.040 ± 0.004	-0.013 ± 0.006
	random uniform	-0.026 ± 0.008	-0.011 ± 0.003	0.013 ± 0.003	-0.028 ± 0.007
	user item average	0.046 ± 0.005	-0.002 ± 0.001	0.051 ± 0.004	-0.003 ± 0.001
Artificial activity	MF	-0.012 ± 0.005	-0.012 ± 0.001	0.002 ± 0.004	-0.002 ± 0.003
	random uniform	-0.031 ± 0.005	-0.021 ± 0.002	-0.003 ± 0.002	-0.007 ± 0.005
	user item average	0.001 ± 0.005	-0.001 ± 0.001	0.003 ± 0.006	-0.001 ± 0.000
Artificial eccentricity	MF	0.138 ± 0.005	0.001 ± 0.000	0.129 ± 0.002	0.008 ± 0.004
	random uniform	0.049 ± 0.007	-0.002 ± 0.001	0.075 ± 0.001	-0.024 ± 0.007
	user item average	0.176 ± 0.004	0.000 ± 0.000	0.176 ± 0.003	-0.001 ± 0.000

For gender, the MF global EO is 0.045. The activity contribution is 0.005, the eccentricity contribution is 0.015, and the other contribution is 0.026. All three contributions therefore have the same sign and reinforce the observed advantage of group A. The residual contribution is the largest of the three, so the gender gap includes substantial structure beyond the two measured distributional factors.

Age exhibits a different decomposition. Its MF global EO is 0.018, with an eccentricity contribution of 0.040, an activity contribution of -0.009, and an other contribution of -0.013. Eccentricity therefore drives the positive overall gap favoring group A, while activity and the remaining factors offset part of it, favoring group B. The gender and age experiments consequently produce similar aggregate quantities at very different component profiles: gender contributions reinforce one another, whereas age contributions partly cancel.

The controlled constructions recover the corresponding hidden factors. For the induced activity construction, the MF overall EO is -0.012 under the unweighted reference distribution, with activity contributing -0.012, eccentricity 0.002, and other factors -0.002. For induced eccentricity, the global EO is 0.138, of which eccentricity contributes 0.129, activity 0.001, and other factors 0.008. The random and user/item-average predictors show related but distinct profiles, confirming that the relative contribution of a factor depends on the predictor as well as on the group distributions.

These results describe practical situations in which two groups can receive different errors. A gap can arise when activity or eccentricity distributions reinforce group membership, when those factors offset one another, or when a substantial residual remains after both are aligned. A fairness audit for demographic groups, user cohorts, or other operational partitions can therefore

report the aggregate EO together with its factor contributions. This indicates whether the disparity is associated with observable interaction patterns, with opposing distributional effects, or with representation and other group-specific structure requiring further investigation.

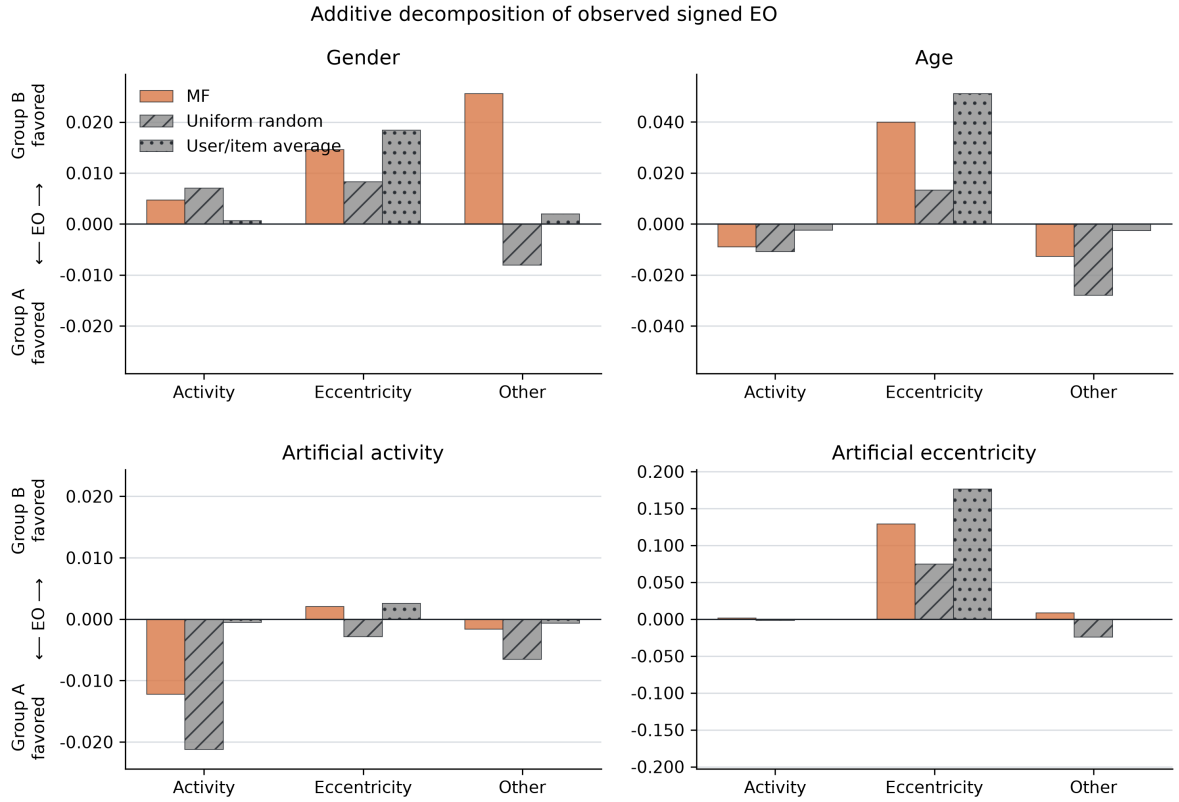


Figure 2: Mean additive signed-EO components across the four group constructions and three predictors.

5.2 Training-distribution intervention

The train-side analysis examines how changing the interaction population affects the learned model. It compares the fully balanced condition $\mathcal{T}_b$ with the matched-size resized control $\mathcal{T}_r$ under the common test distribution defined by the test-side weighting procedure.

Table 2: Training-distribution intervention. Values are means across folds.

Condition	EO	RMSE A	RMSE B
$\mathcal{T}_b$ (Fully balanced)	0.023	0.871	0.894
$\mathcal{T}_r$ (Resized control)	0.042	0.850	0.892

The balanced training population produces EO 0.023, compared with 0.042 for the resized control. Its RMSE for group A is 0.871 versus 0.850, while group B remains similar at 0.894 versus 0.892. The result links training population composition to the learned model's group error profile and shows that this effect persists when the two training conditions have the same rating count.

6 Conclusion

Global group fairness metrics alone are insufficient to interpret group error in collaborative filtering: unequal interaction distributions can produce, reinforce, or conceal a disparity. We proposed a

distribution-aware decomposition that addresses this gap with counterfactual alignment of rating behaviour distributions on the evaluation side and isolation of training-population effects on the learning side, without using group membership as a predictor feature.

Our empirical evidence, employing our group unfairness decomposition mechanism, indicates that the same apparent gap can have different internal profiles. In the different demographic constructions in movie prediction tasks, our method was able to discern which of the overall observed bias was caused, concealed or offset by eccentricity, activity or other psychosiciological factors being disparate between groups. Balancing population composition likewise altered the group error gap at matched size, with unequal consequences for the two groups.

Our findings suggest that researchers and practitioners should report group unfairness as a sum of different distributional factor contributions and residual structure alongside aggregate bias, rather than treating the overall bias as evidence of a single unfairness mechanism.

Further work should examine additional rating prediction tasks and extend the analysis to multisided fairness. The scope of the evidence is limited to movie recommendation due to the unavailability of other public datasets.

A Group-wise RMSE decomposition

The EO decomposition in Section 3.2.1 can also be applied separately to the RMSE of each group. The table below reports the unweighted starting RMSE and the additive activity, eccentricity, and remaining RMSE contributions for each construction, predictor, and group. The remaining RMSE term represents the portion not explained by the measured activity or eccentricity factors. The three contributions sum to the starting RMSE; a positive contribution increases the reconstructed starting error, whereas a negative contribution offsets it. Positive contributions are prefixed with a plus sign in the table.

Table 3: Group-wise RMSE decomposition across ten folds. The starting RMSE is decomposed into activity, eccentricity, and remaining RMSE contributions using the same Shapley construction as the EO decomposition. Positive contributions indicate increases in RMSE; negative contributions indicate decreases. Values are means ± standard deviations.

Construction	Predictor	Group	Starting RMSE	Activity	Eccentricity	Remaining RMSE
Gender	MF	A	0.833 ± 0.002	-0.003 ± 0.002	-0.004 ± 0.002	0.840 ± 0.003
		B	0.878 ± 0.005	+0.001 ± 0.002	+0.011 ± 0.003	0.865 ± 0.004
	random uniform	A	1.706 ± 0.004	-0.005 ± 0.001	-0.002 ± 0.002	1.713 ± 0.005
		B	1.714 ± 0.007	+0.002 ± 0.002	+0.006 ± 0.002	1.705 ± 0.008
	user/item average	A	0.951 ± 0.002	-0.000 ± 0.002	-0.004 ± 0.002	0.956 ± 0.004
		B	0.972 ± 0.006	+0.001 ± 0.002	+0.014 ± 0.004	0.958 ± 0.004
Age	MF	A	0.828 ± 0.007	+0.003 ± 0.002	-0.019 ± 0.002	0.844 ± 0.006
		B	0.846 ± 0.002	-0.006 ± 0.001	+0.021 ± 0.002	0.831 ± 0.003
	random uniform	A	1.731 ± 0.007	+0.000 ± 0.002	-0.002 ± 0.002	1.733 ± 0.007
		B	1.705 ± 0.004	-0.010 ± 0.001	+0.011 ± 0.002	1.705 ± 0.005
	user/item average	A	0.916 ± 0.005	+0.002 ± 0.001	-0.021 ± 0.002	0.934 ± 0.004
		B	0.962 ± 0.002	-0.000 ± 0.001	+0.030 ± 0.003	0.932 ± 0.004
Artificial activity	MF	A	0.850 ± 0.004	+0.006 ± 0.001	-0.002 ± 0.003	0.846 ± 0.004
		B	0.838 ± 0.003	-0.007 ± 0.002	+0.001 ± 0.002	0.844 ± 0.003
	random uniform	A	1.723 ± 0.004	+0.007 ± 0.002	+0.001 ± 0.001	1.715 ± 0.005
		B	1.693 ± 0.006	-0.014 ± 0.001	-0.002 ± 0.001	1.708 ± 0.006
	user/item average	A	0.956 ± 0.004	+0.000 ± 0.002	-0.001 ± 0.003	0.957 ± 0.004
		B	0.957 ± 0.003	-0.000 ± 0.003	+0.001 ± 0.003	0.956 ± 0.004
Artificial eccentricity	MF	A	0.772 ± 0.004	-0.007 ± 0.001	-0.036 ± 0.001	0.814 ± 0.004
		B	0.910 ± 0.002	-0.005 ± 0.001	+0.093 ± 0.003	0.822 ± 0.003
	random uniform	A	1.683 ± 0.004	-0.003 ± 0.001	-0.019 ± 0.001	1.705 ± 0.005
		B	1.732 ± 0.006	-0.005 ± 0.001	+0.056 ± 0.001	1.681 ± 0.006
	user/item average	A	0.864 ± 0.003	-0.007 ± 0.001	-0.048 ± 0.001	0.919 ± 0.003
		B	1.040 ± 0.003	-0.007 ± 0.001	+0.129 ± 0.003	0.919 ± 0.003

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